VPA IB DP CS — Year 1 Semester 2 Assessment Pack

Year 1 Semester 2 Assessment Pack 评估包 Pínggū Bāo

Topics: 3 (Networks) + 4 (Computational Thinking) | Level: SL + HL

1. Assessment Overview

This assessment pack covers Topic 3 (Networks) and Topic 4 (Computational Thinking). It includes Paper 1 and Paper 2 style questions, IA milestone assessments, and markschemes aligned to IB standards.

2. Assessment Calendar

Week	Assessment	Type	Duration	Marks
5	Topic 3 Progress Check	Formative	40 min	30
9	Topic 3 End-of-Topic Test	Summative	1 hour	50
13	Programming Progress Check	Formative	1 hour	40
17	Topic 4 End-of-Topic Test	Summative	1h30	60 (SL) / 65 (HL)
18	Semester 2 Examination	Summative	2 hours	100
8	IA Design Document	IA Milestone	Homework	Criterion B

3. Topic 3 Sample Questions

Q1 — Define [2]

(a) Define the term packet switching. (分组交换 fēnzǔ jiāohuàn) [1]

(b) Define the term protocol. (协议 xiéyì) [1]

Q2 — Describe [4]

Describe the function of a router in a network.

Award [1] for: forwards packets between networks; [1] for: reads destination IP; [1] for: uses routing table; [1] for: chooses best path.

Q3 — Explain [6]

Explain how a firewall protects a network from external threats.

Q4 — Compare [6]

Compare client-server and peer-to-peer network architectures.

Q5 — Evaluate [10]

A multinational corporation is considering migrating its file servers to a public cloud provider. Evaluate this decision from the perspectives of cost, security, and reliability.

4. Topic 4 Sample Questions

Q6 — Trace [4]

Trace the following algorithm using a trace table, showing the values of x, y, and z at each iteration:

x ← 5

y ← 1

while x > 0 do

z ← x * y

y ← y + 1

x ← x □ 2

end while

Q7 — Construct [6]

A teacher needs to find the highest score in an array of 50 test scores.

(a) Construct pseudocode for a linear search to find the maximum value. [4]

(b) Identify one piece of test data you would use to test this algorithm, stating whether it is normal, boundary, or erroneous. [2]

Q8 — Explain [5]

Explain why binary search is more efficient than linear search for finding an item in a sorted array of one million elements.

Q9 — Analyse [8]

Analyse the efficiency of bubble sort and merge sort, comparing their best-case, average-case, and worst-case time complexities.

Q10 — Discuss [12]

Discuss the importance of thorough testing in software development, with reference to at least two real-world software failures.

5. Paper 2 Programming Exercise Format

Paper 2 questions require students to write pseudocode solutions. Markschemes award marks for:

Correct algorithmic approach

Appropriate use of data structures

Correct syntax (IB pseudocode conventions)

Input validation where appropriate

Meaningful variable names and internal commentary

6. IA Criterion B — Design Document Rubric

Level	Descriptor
0	No evidence of design
1 - 2	Basic record of tasks; minimal design documentation
3 - 4	Clear record of tasks; system overview diagram; some algorithm design
5 - 6	Detailed record of tasks with dates; comprehensive system design with diagrams; well-designed algorithms with test plans

7. Bilingual Command Terms for Assessments

Command Term	中文	Pinyin	What Students Must Do
Define	定义	dìngyì	Give precise meaning
Describe	描述	miáoshù	Give a detailed account
Explain	解释	jiěshì	Give reasons/causes
Compare	比较	bǐjiào	Give similarities and differences
Evaluate	评估	pínggū	Assess strengths/limitations with judgement
Discuss	讨论	tǎolùn	Offer balanced review with conclusion
Construct	构建	gòujiàn	Build/create from components
Analyse	分析	fēnxī	Break down into components
Trace	追踪	zhuīzōng	Follow step by step

☐ Victoria Park Academy Shenzhen | IB DP Computer Science | Assessment Pack